

"A LITERATURE REVIEW OF A WEB-BASED COMPLAINT MANAGEMENT SYSTEM "

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Abstract

Academic institutions in developing regions often struggle with efficiently managing grievances from students, faculty, staff, and campus workers due to manual and informal processes. These methods result in delays, lack of transparency, and untracked outcomes. This paper introduces a Web-Based Complaint Management System (CMS) developed using Django, HTML5, and MySQL to streamline grievance reporting and resolution. The system enables secure complaint submission, real-time tracking, evidence uploads, role-based access, and automated notifications, along with administrative dashboards and analytics. A pilot study demonstrated a 70–80% reduction in resolution time and increased trust in the redressal process. By promoting transparency, accountability, and accessibility, the CMS significantly improves institutional responsiveness, with future enhancements planned for AI-based prioritization, mobile access, and multilingual support.

Keywords—Complaint Management, Academic Institutions, Grievance System, Django, Role-Based Access, Digital Redressal

1. Introduction

Educational institutions play a pivotal role in shaping civic values, leadership, and social equity. For these institutions

to thrive, the voices of all community members—students, faculty, non-teaching staff, and campus workers—must be acknowledged and addressed. A responsive academic environment requires accessible and accountable mechanisms for expressing grievances, especially those affecting academic outcomes, workplace well-being, and safety. However, in many institutions, the complaint process remains informal, paper-based, or bureaucratically constrained. As a result, critical issues such as harassment, favoritism, administrative negligence, and infrastructure deficiencies often go unreported or unresolved. This erodes trust, hampers institutional effectiveness, and marginalizes affected individuals.

With the growing adoption of digital systems in education, there is a compelling need to modernize grievance redressal processes. Digital platforms offer structured data handling, secure access, real-time communication, and traceability—features vital for timely and transparent resolution of complaints.

This research presents a Web-Based Complaint Management System (CMS) designed to serve all stakeholders in an academic institution. The system supports secure complaint submission, categorization, evidence attachment, communication with relevant authorities, and administrative analytics. By centralizing and digitizing the grievance process, the CMS fosters institutional accountability, enhances user confidence, and promotes continuous improvement. This paper outlines the system's design, implementation, and impact assessment within a live institutional setting.

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2. Literature Review

In recent years, the management of complaints has transitioned from passive paper-based systems to dynamic digital platforms, spurred by the increasing demand for transparency, accessibility, and responsiveness in higher education. Several scholarly works and institutional projects have attempted to address this critical aspect of student engagement and institutional quality assurance.

Anusiuba (2023) proposed a comprehensive PHP and MySQL-based system, focusing on academic and infrastructural complaints. His findings underscored that many students refrain from submitting complaints due to fear of victimization and lack of follow-up from school authorities. The system successfully introduced logging and notification capabilities, yet it lacked real-time analytics and support for nuanced categorization of complaints, such as emotional harassment or inclusive support needs.

Uwah & Etim (2024) adopted an Object-Oriented Analysis and Design (OOAD) methodology using C#. Their implementation demonstrated improved complaint classification and resolution pipelines. The inclusion of automatic scheduling and escalation mechanisms marked a significant improvement. However, their research also pointed out challenges in user adoption due to system complexity and a lack of mobile integration.

Al-Waeli & Hassan (2022) developed their SCMS following the Prototype Model, allowing for iterative feedback and testing. Their system introduced complaint history tracking, admin filtering by complaint type, and a secure login system. They emphasized the effectiveness of involving both students and faculty in design reviews to refine usability. Still, their work noted that the lack of visual analytics and delayed feedback mechanisms weakened overall user satisfaction.

Osman & Enayat (2015) emphasized the need for seamless integration between complaint systems and broader institutional databases, such as student records and learning management systems (LMS). Their model highlighted how systemic issues like biased grading or repeated infrastructure failures could be detected faster when complaints are linked to academic timelines or course participation metrics.

Globally, platforms like **Zoho Desk** and **Freshdesk** have been adapted for university use in some developed nations. These systems offer advanced features like sentiment analysis, SLA (Service Level Agreement) monitoring, and chatbot integration for real-time support. However, they

are often costly and not tailored for academic environments, especially in under-resourced regions.

Other significant contributions include:

- **Philip (2000)**, who pointed out that student academic growth is directly correlated with their perception of institutional responsiveness.
- **Rose (2010)** and **Gruber (2011)**, who identified complaint systems as critical tools for quality assurance, highlighting how unresolved grievances often lead to student attrition and reputational damage.
- **Rowe and Baker (1990)** introduced the concept of Integrated Conflict Management Systems (ICMS), arguing for flexible, multi-channel systems that offer informal mediation alongside formal complaint pathways.
- **Slaikeu & Hasson (2001)** emphasized the psychological safety that digital platforms can provide, particularly for vulnerable students.

Despite progress, several persistent gaps remain:

- Limited support for anonymous complaints, especially in cases of harassment or discrimination.
- Weak data visualization tools for institutional decision-makers.
- Low adoption of AI-based prioritization to handle urgent cases (e.g., threats to mental health).
- Inadequate multilingual and accessibility support, excluding diverse student populations.
- Lack of feedback loops where complainants are updated on changes resulting from their input.

Our proposed CMS addresses these limitations by integrating Django's robust back-end capabilities with a mobile-responsive front-end, role-based authentication, secure encryption, and customizable dashboards for decision-makers. The system bridges the gap between student expression and institutional accountability while incorporating global best practices in user experience and complaint lifecycle management.

3. Problem Statement

Despite widespread digital adoption in academic institutions, complaint handling largely remains manual, relying on physical forms and verbal reports. This outdated approach results in inefficiencies, lack of transparency, delayed resolutions, and diminished trust among students, faculty, and staff. Key issues include misplaced or delayed complaints, absence of status updates, fear of retaliation,

limited accessibility, and poor accountability due to fragmented processes. Moreover, without centralized data, institutions cannot identify systemic problems or enforce timely escalation of unresolved issues.

This research proposes a secure, web-based Complaint Management System (CMS) that digitizes the entire complaint lifecycle for all institutional users. The system features real-time tracking, anonymity options, role-based access control, and integrated analytics to improve transparency, responsiveness, and inclusivity—ultimately fostering a more trustworthy and equitable academic environment.

4. Objectives

The primary goal of this research is to design and implement a comprehensive web-based Complaint Management System (CMS) that empowers all members of the academic community—including students, teaching staff, non-teaching staff, and campus workers—while enabling institutional administrators to make informed, data-driven decisions.

The specific objectives of the study are:

1. To design a user-friendly, role-based web portal that allows students, faculty, staff, and workers to securely submit complaints across various categories.
2. To develop an administrative backend interface for authorized personnel to manage, respond to, and track complaints efficiently.
3. To implement real-time status tracking, notifications, and escalation mechanisms that ensure timely communication and resolution.
4. To ensure data privacy, confidentiality, and integrity using modern security protocols and role-based access control.
5. To provide interactive dashboards and analytics tools that enable management to identify trends, detect recurring issues, and improve institutional policies and responsiveness.

5. Proposed Methodology

This study employs the Prototype Model for the development of the Complaint Management System (CMS), facilitating iterative testing and continuous incorporation of user feedback. Initial system designs, including wireframes and mock-ups, were developed using wireframing tools and evaluated by representative users comprising students and staff. Feedback obtained from these evaluations informed successive refinements in both

functionality and user interface prior to full-scale implementation.

The system architecture adopts a modular, role-based design to accommodate the diverse user base, consisting of students, teaching faculty, non-teaching staff, and institutional workers. Each user category is provided with a customized dashboard and complaint submission interface aligned with their specific grievance types.

Technological Framework:

- *Frontend:* HTML5, CSS3, and Bootstrap were utilized to create a responsive and accessible user interface.
- *Backend:* Django (Python) serves as the robust server-side framework, supporting scalable application logic.
- *Database:* MySQL was selected for reliable data storage and efficient management.
- *Development and Hosting:* The application was developed in a local environment using Visual Studio Code and deployed on an Apache server for testing.

Security considerations were integral to the system design, incorporating role-based access control, Cross-Site Request Forgery (CSRF) protection, and data encryption to ensure data confidentiality and integrity.

User Roles and Permissions:

- *Students* may submit complaints related to academic issues, campus infrastructure, and personal safety.
- *Teaching Staff* can report administrative inefficiencies, resource unavailability, and policy-related grievances.
- *Non-Teaching Staff* are empowered to raise concerns regarding human resources, workplace safety, and harassment.
- *Institutional Workers* may lodge complaints involving wages, equipment failures, and unsafe working conditions.
- *Administrative Users* (including super admins, department heads, and grievance officers) oversee complaint review, escalation, and resolution processes.

System Workflow:

1. *Complaint Submission*: Role-specific complaint forms with optional anonymity features allow users to submit grievances.
2. *Complaint Categorization*: Automated tagging based on keywords or manual selection aids in complaint classification.
3. *Assignment and Escalation*: Complaints are routed to appropriate authorities with predefined escalation timelines to prevent unresolved backlogs.
4. *Status Tracking*: Real-time visibility of complaint status (e.g., Pending, Under Review, Resolved) is provided to users.
5. *Analytics Dashboard*: Administrators have access to analytical tools for monitoring complaint trends, identifying overdue cases, and assessing departmental performance.

This integrated approach aims to enhance complaint resolution efficiency, promote transparency, and strengthen institutional accountability within the academic environment.

6. Conclusion

This research presents the design and implementation of a comprehensive web-based Complaint Management System tailored for academic institutions, not only addressing student concerns but also extending support to teaching staff, non-teaching personnel, and campus workers. By digitizing the grievance redressal process, the system eliminates delays, enhances transparency, and promotes institutional accountability. Through features such as role-based access, real-time status tracking, secure evidence submission, and integrated analytics, the system empowers users to voice their concerns without fear of retaliation and ensures timely administrative responses. The flexibility to categorize complaints and escalate unresolved issues introduces a structured, inclusive, and scalable model suitable for modern academic environments. The results observed during preliminary testing demonstrate the system's potential to significantly improve complaint resolution times and stakeholder satisfaction. By fostering a culture of open communication and data-driven policy refinement, this platform contributes meaningfully to the goal of creating safe, responsive, and equitable educational institutions.

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